

Tutorial -7-

Quick Revision

Knowledge Representation & Reasoning

Resolution: Inference procedure

Inference procedures based on resolution work by using the principle of proof by contradiction:

To show that $KB \models \alpha$ we show that $(KB \wedge \neg\alpha)$ is unsatisfiable

The process: 1. convert $KB \wedge \neg\alpha$ to CNF
2. resolution rule is applied to the resulting clauses.

<u>Modus Ponens</u>	$\frac{\alpha \Rightarrow \beta, \quad \alpha \quad \text{Premise}}{\beta \quad \text{Conclusion}}$
<u>Modus Tollens</u>	$\frac{\alpha \Rightarrow \beta, \quad \neg \beta \quad \text{Premise}}{\neg \alpha \quad \text{Conclusion}}$
<u>AND elimination</u>	$\frac{\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_n \quad \text{Premise}}{\alpha_i \quad \text{Conclusion}}$
And-Introduction	$\frac{\alpha_1, \alpha_2, \dots, \alpha_n \quad \text{Premise}}{\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_n \quad \text{Conclusion}}$
Double Negation	$\frac{\neg \neg \alpha \quad \text{Premise}}{\alpha \quad \text{Conclusion}}$
Full resolution inference rule:	$\frac{l_1 \vee \dots \vee \underline{l_k} \quad m_1 \vee \dots \vee \underline{m_n}}{l_1 \vee \dots \vee l_{i-1} \vee l_{i+1} \vee \dots \vee \underline{l_k} \vee m_1 \vee \dots \vee m_{j-1} \vee m_{j+1} \vee \dots \vee \underline{m_n}}$

Tutorial

Q1: Proof by resolution (AIMA 3rd Ed. Page 252)

a) Convert into Conjunctive Normal Form (CNF) the sentence $(KB \wedge \neg \alpha)$ given :

$$KB: (B_{1,1} \Leftrightarrow P_{1,2} \vee P_{2,1}) \wedge \neg B_{1,1}$$

$$\alpha: \neg P_{1,2}$$

• Sentence: $(B_{11} \Leftrightarrow P_{12} \vee P_{21}) \wedge \neg B_{11} \wedge \neg P_{12} \quad // (KB \wedge \neg \alpha)$

• Solution:

• Using bidirectional elimination,

$$(B_{11} \Rightarrow (P_{12} \vee P_{21})) \wedge ((P_{12} \vee P_{21}) \Rightarrow B_{11}) \wedge \neg B_{11} \wedge \neg P_{12}$$

• Using Implication elimination,

$$(\neg B_{11} \vee (P_{12} \vee P_{21})) \wedge (\neg(P_{12} \vee P_{21}) \vee B_{11}) \wedge \neg B_{11} \wedge \neg P_{12}$$

• De Morgan (Not),

$$(\neg B_{11} \vee P_{12} \vee P_{21}) \wedge ((\neg P_{12} \wedge \neg P_{21}) \vee B_{11}) \wedge \neg B_{11} \wedge \neg P_{12}$$

• Distribution of $\vee$ over $\wedge$

$$(\neg B_{11} \vee P_{12} \vee P_{21}) \wedge (\neg P_{12} \vee B_{11}) \wedge (\neg P_{21} \vee B_{11}) \wedge \neg B_{11} \wedge \neg P_{12}$$

• Double negation elimination

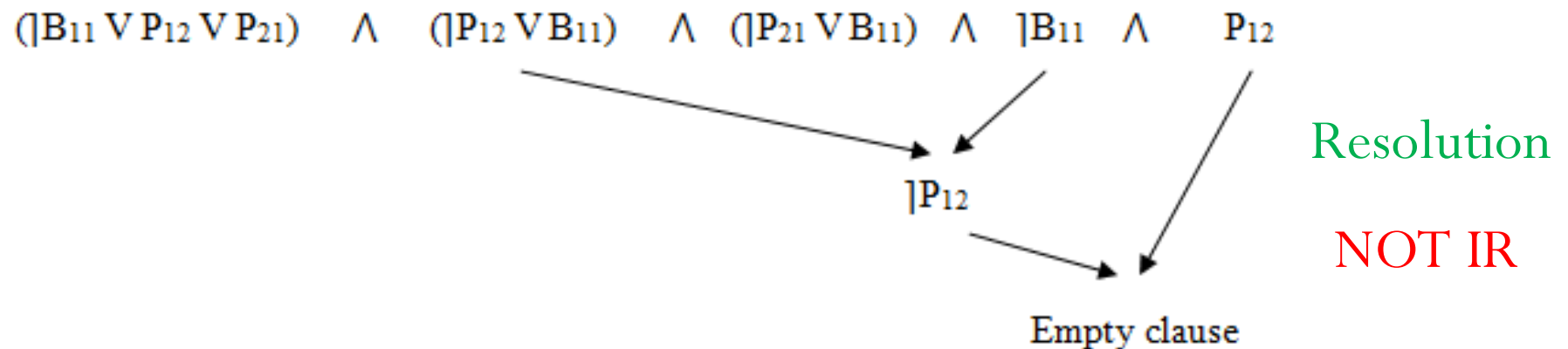
$$(\neg B_{11} \vee P_{12} \vee P_{21}) \wedge (\neg P_{12} \vee B_{11}) \wedge (\neg P_{21} \vee B_{11}) \wedge \neg B_{11} \wedge P_{12}$$

$$(CNF): (\neg B_{11} \vee P_{12} \vee P_{21}) \wedge (\neg P_{12} \vee B_{11}) \wedge (\neg P_{21} \vee B_{11}) \wedge \neg B_{11} \wedge P_{12}$$

Q1: Proof by resolution (AIMA 3rd Ed. Page 252)

b) Using the **CNF** result from (a), show that the knowledge base (KB) entails the sentence α , i.e. $KB \models \alpha$. Which principle did you use to prove the entailment?

- (AIMA page 255) Using principle of proof by contradiction, to show that $KB \models \alpha$, we show that $(KB \wedge \neg \alpha)$ is unsatisfiable



$\therefore KB \wedge \neg \alpha$ is unsatisfiable

$\therefore KB \models \alpha$

Q2: Use the PL inference rules and the following PL sentences to prove P

1. $R \vee T$
2. $Q \wedge U$
3. $(S \vee T) \Rightarrow (Q \Rightarrow P)$
4. $\neg R \vee S$

● **Solution:**

- R1, R4 applying Resolution Rule
 - R5: $S \vee T$ $\frac{R \vee T, \neg R \vee S}{T \vee S}$
- R3, R5 applying modus ponens
 - R6: $Q \Rightarrow P$
- R2 applying and-elimination
 - F1: Q
- F1, R6 applying modus ponens
 - Conclusion: P

Q3: Translate facts into prepositional logic:

1. A is on the right of B:

$$R_{AB}$$

2. B is on the left of A:

$$L_{BA}$$

3. C is on the top of B:

$$T_{cB}$$

4. C is on the top of A:

$$T_{cA}$$

5. If A is on the right of B, then B is on the left of A:

$$R1: R_{AB} \Rightarrow L_{BA}$$

6. If C is on the top of B which is on the left of A, then it is not on the top of A.

$$R2: T_{cB} \wedge L_{BA} \Rightarrow \neg T_{cA}$$

Then if you know that in such world, A is on the right of B and C on the top of B,
Can you deduce from the previous facts that C is not on the top of A.

Facts:

R_{AB}

T_{CB}

Rule

$R_{AB} \wedge T_{CB} \Rightarrow \neg T_{CA}$

Modus ponens: $\neg T_{CA}$

- Rules

$R1: R_{AB} \Rightarrow L_{BA}$, $R2: T_{CB} \wedge L_{BA} \Rightarrow \neg T_{CA}$

- Fact (knowing that R_{AB} and T_{CB})

$F1: R_{AB}$, $F2: T_{CB}$

- Prove ($\neg T_{CA}$)

$F1$ & $R1$ modus ponens gives $F3: L_{BA}$

$F2$ & $F3$ and-introduction gives $F4: T_{CB} \wedge L_{BA}$

$F4$ & $R2$ modus ponens gives conclusion $\neg T_{CA}$